

Computer Science & IT

Compiler Design



Syntax Directed Translation

Lecture No. 02



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Recap of Previous Lecture



Topic

$LR(0)$ $SLR(1)$

Topic

Topic

Topics to be Covered



Topic

$LR(0), SLR(1)$

Topic

Topic

$LL(1)$ ✓
 $SLR(1)$
 $LR(0)$ ✓

$S \rightarrow dA \{d\}$ ✓
 $\quad \quad \quad | aB \{a\}$ ✓
 $A \rightarrow bA \{b\}$ ✓
 $\quad \quad \quad | c \quad \{c\}$ ✓
 $B \rightarrow bB \{b\}$ ✓
 $\quad \quad \quad | c \quad \{c\}$ ✓

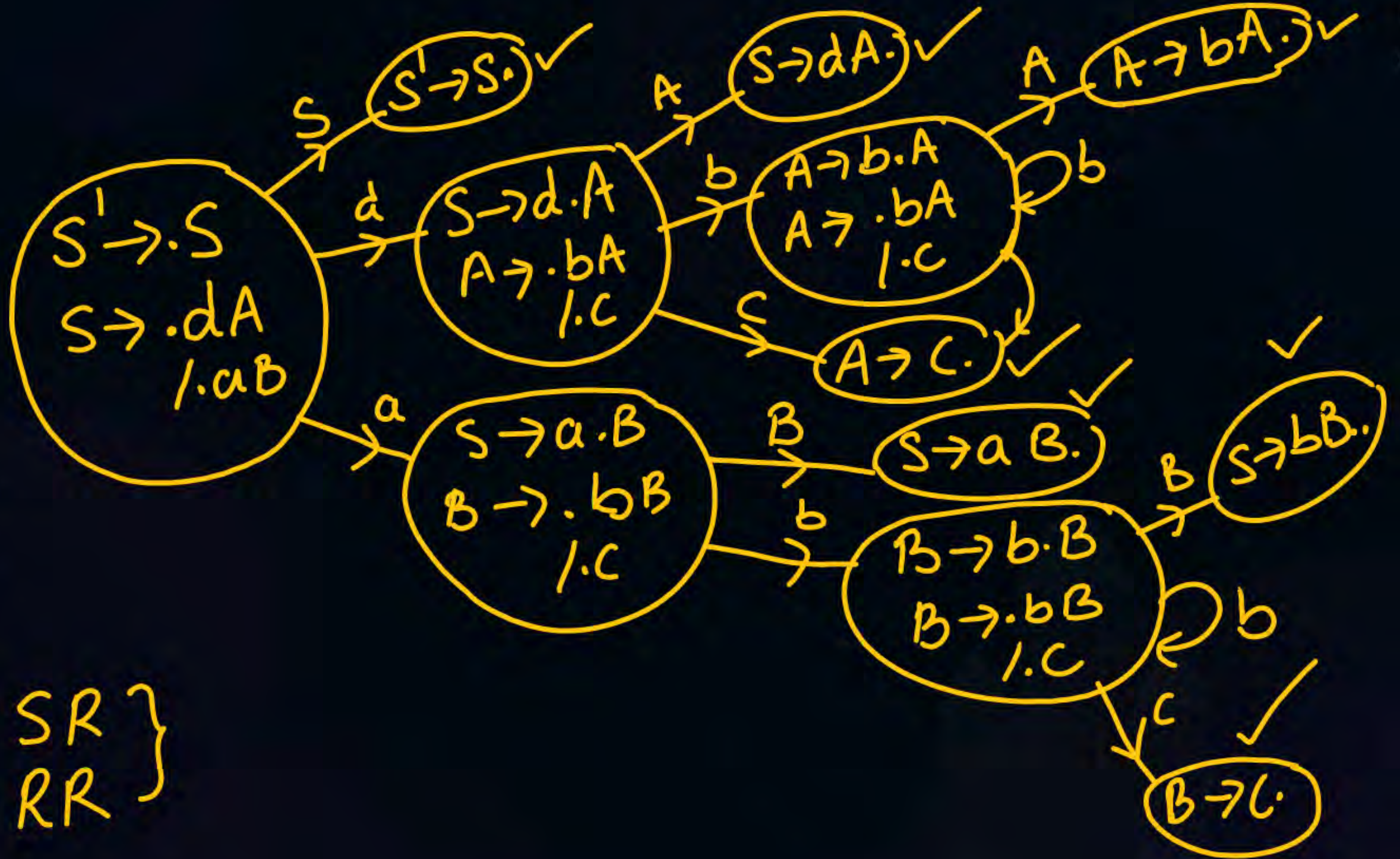
$LL(1)$ table

	a	b	c	d
S	$S \rightarrow aB$			$S \rightarrow dA$
A		$A \rightarrow bA$	$A \rightarrow c$	
B		$B \rightarrow bB$	$B \rightarrow c$	

$LL(1)$

$LL(1) \checkmark$
 $SLR(1) \checkmark$
 $LR(0) \checkmark$

$S \rightarrow dA \{d\}$
 $A \rightarrow bA \{b\}$
 $B \rightarrow bB \{b\}$
 $\{c\}$



$SR \}$
 $RR \}$

LL(1) LR(0) SLR(1)



$S \rightarrow \bar{A}/\bar{a}$
 $A \rightarrow a$

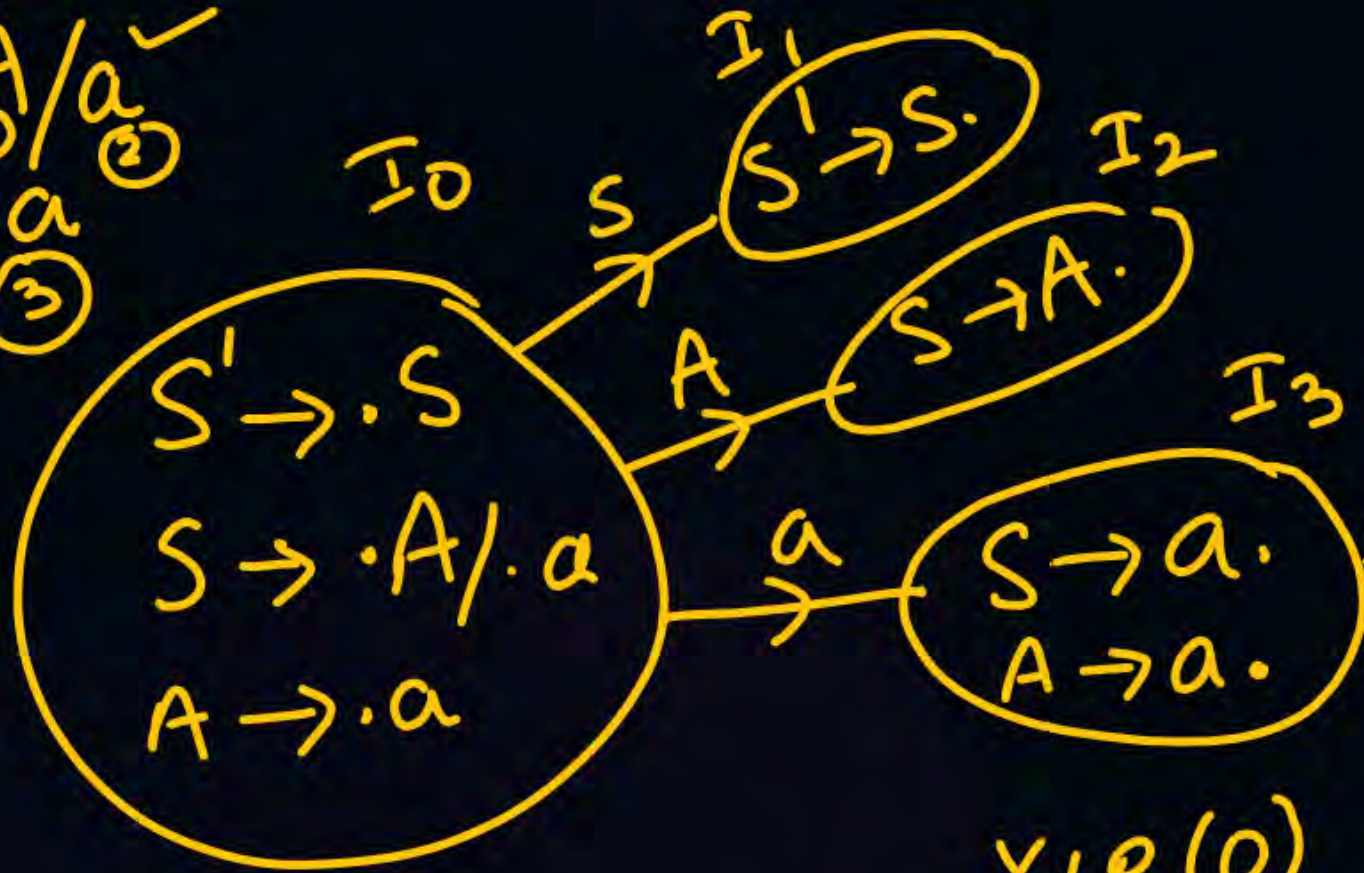
LL(1)

	a	\$
S	$S \rightarrow A / S \rightarrow a$	
A		

Not LL(1)

LL(1) LR(0) SLR(1)

X ✓
 $S \rightarrow A/a$
 $A \rightarrow a$

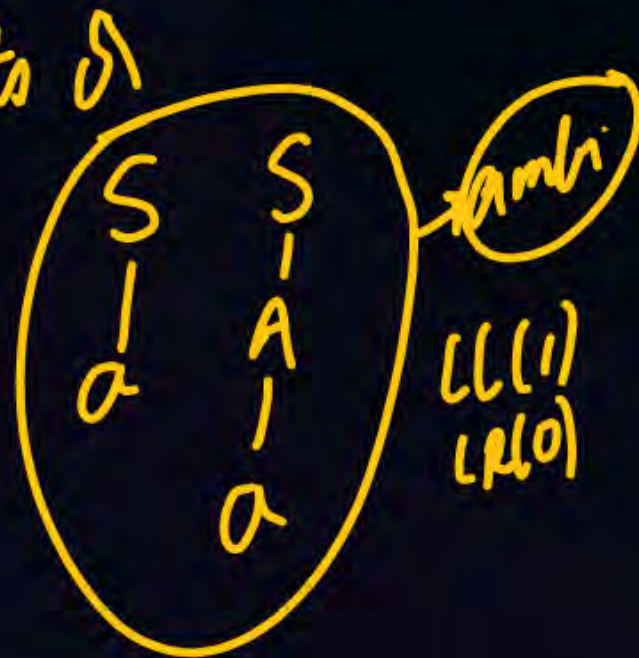


SLR(1)

	a	\$
I ₃	r_2/r_3	

LR(0)

XL R(0)



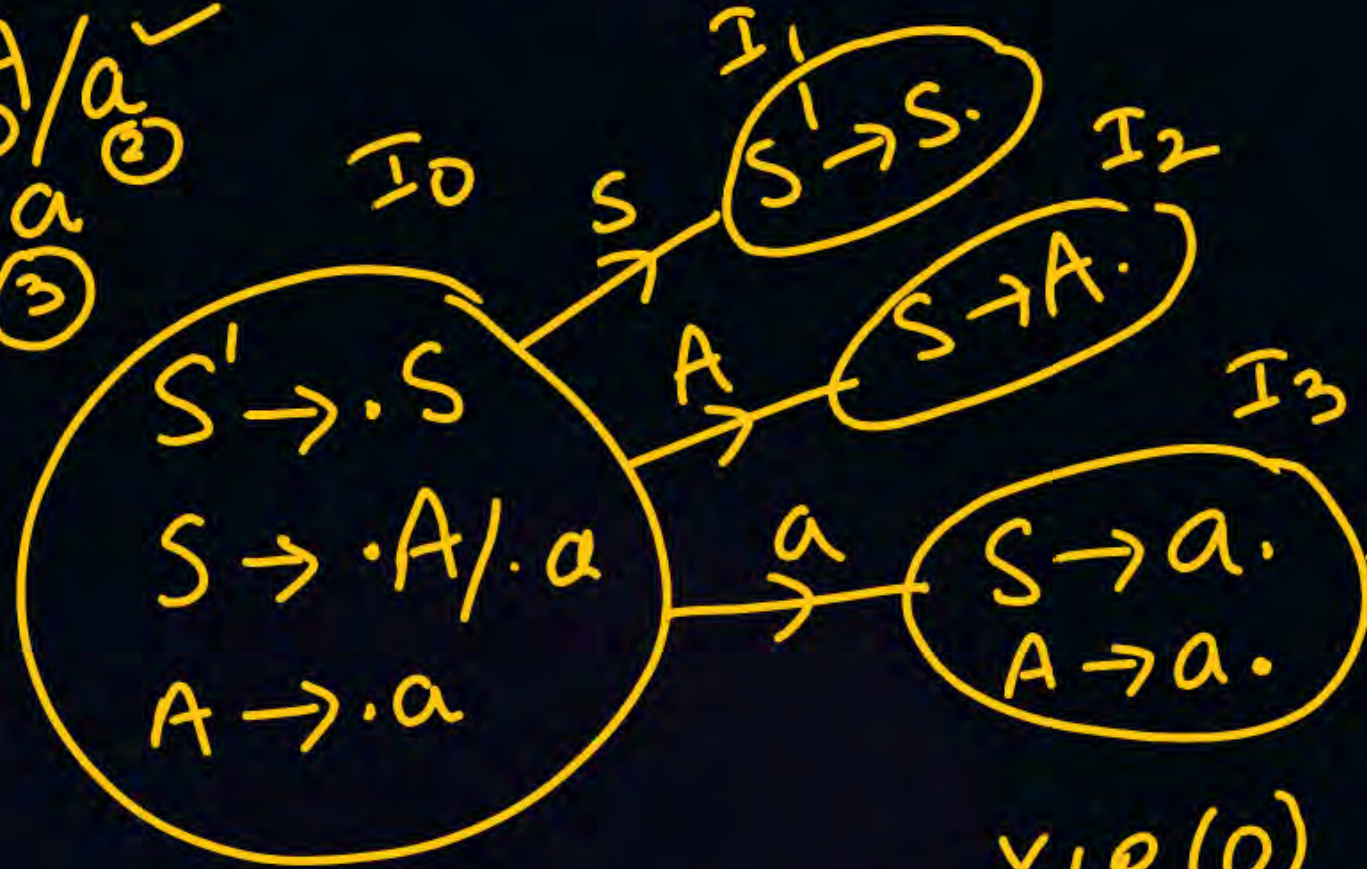
The states with SR Conflicts or RR conflicts are called inadequate states.

	a	\$
I ₃	R_2/R_3	R_2/R_3

LL(1) LR(0) SLR(1)



X ✓
 $S \rightarrow A/a$
 $A \rightarrow a$

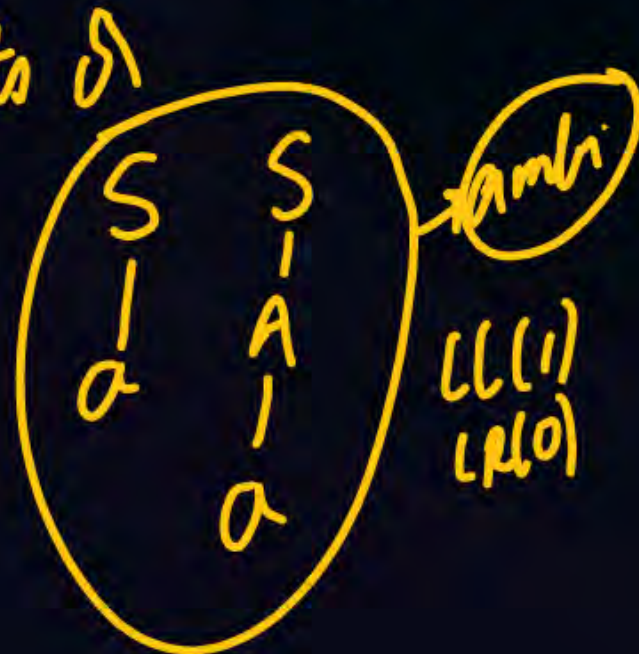


SLR(1)

	a	\$
I3		r2/r3

LR(0)

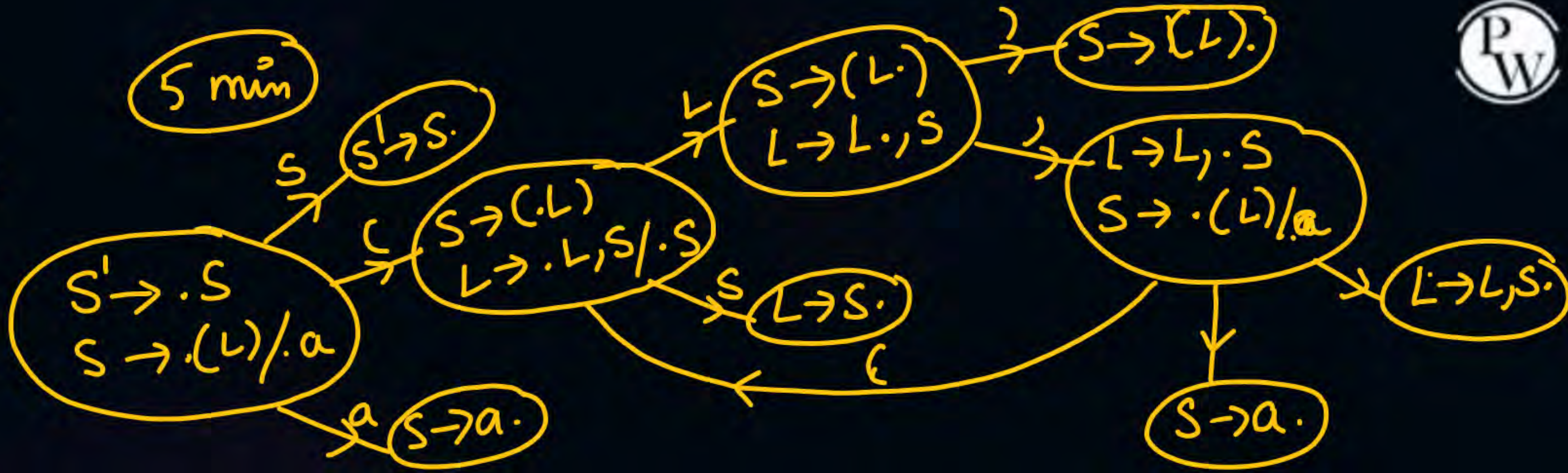
XL R(0)



The states with SR conflicts or
 RR conflicts are called
 inadequate states.

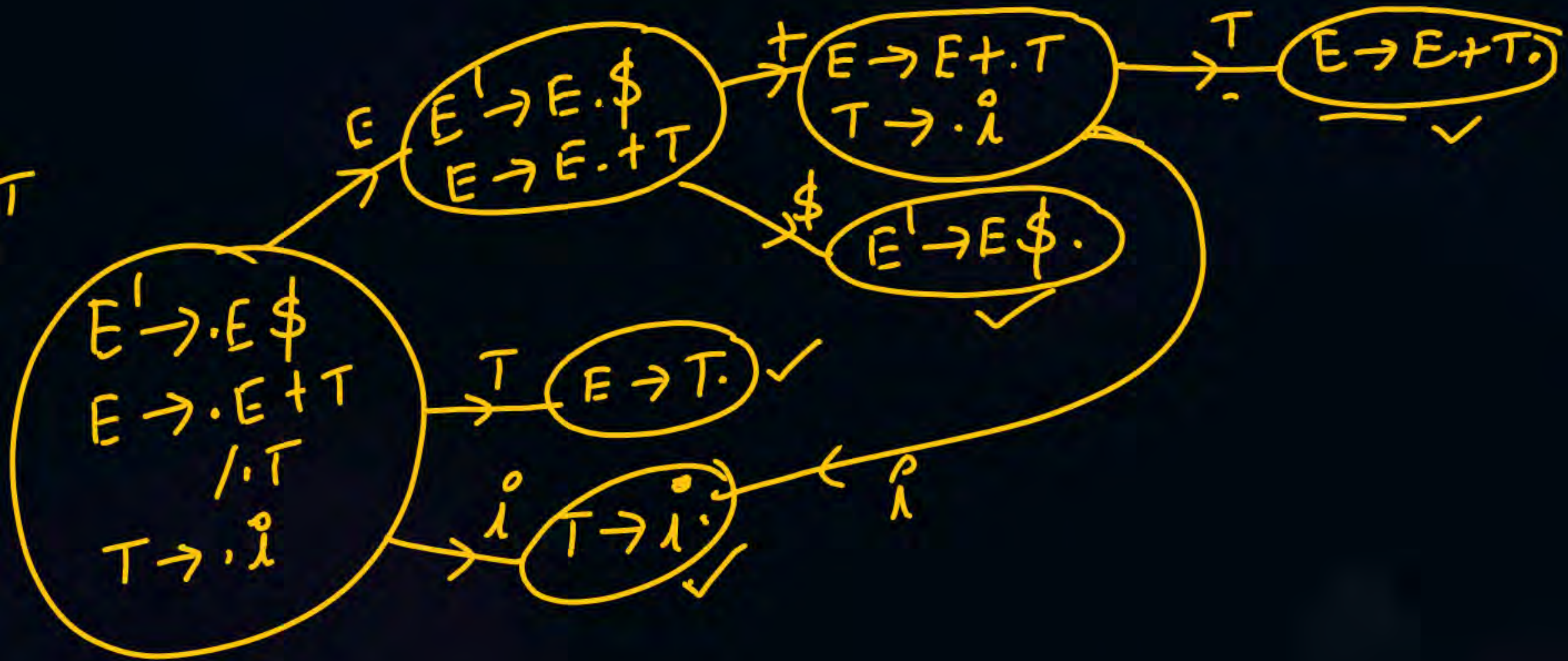
	a	\$
I3	R2/R3	R2/R3

$S \rightarrow (L)/a$
 $L \rightarrow L, S/S$



LL(1) ✗
 LR(0) ✓
 SLR(1) ✓

$E \rightarrow E + T$
 $T \rightarrow i$



$LL(1) \times$
 $LR(0) \checkmark$
 $SLR(1) \checkmark$

Every $LR(0)$ grammar is $SLR(1)$ also

$E \rightarrow T + E$ ①
 $\quad \quad \quad \cdot T$ ②
 $T \rightarrow i$ ③

LL(1)

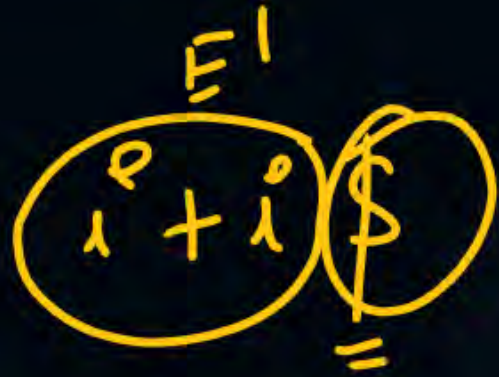
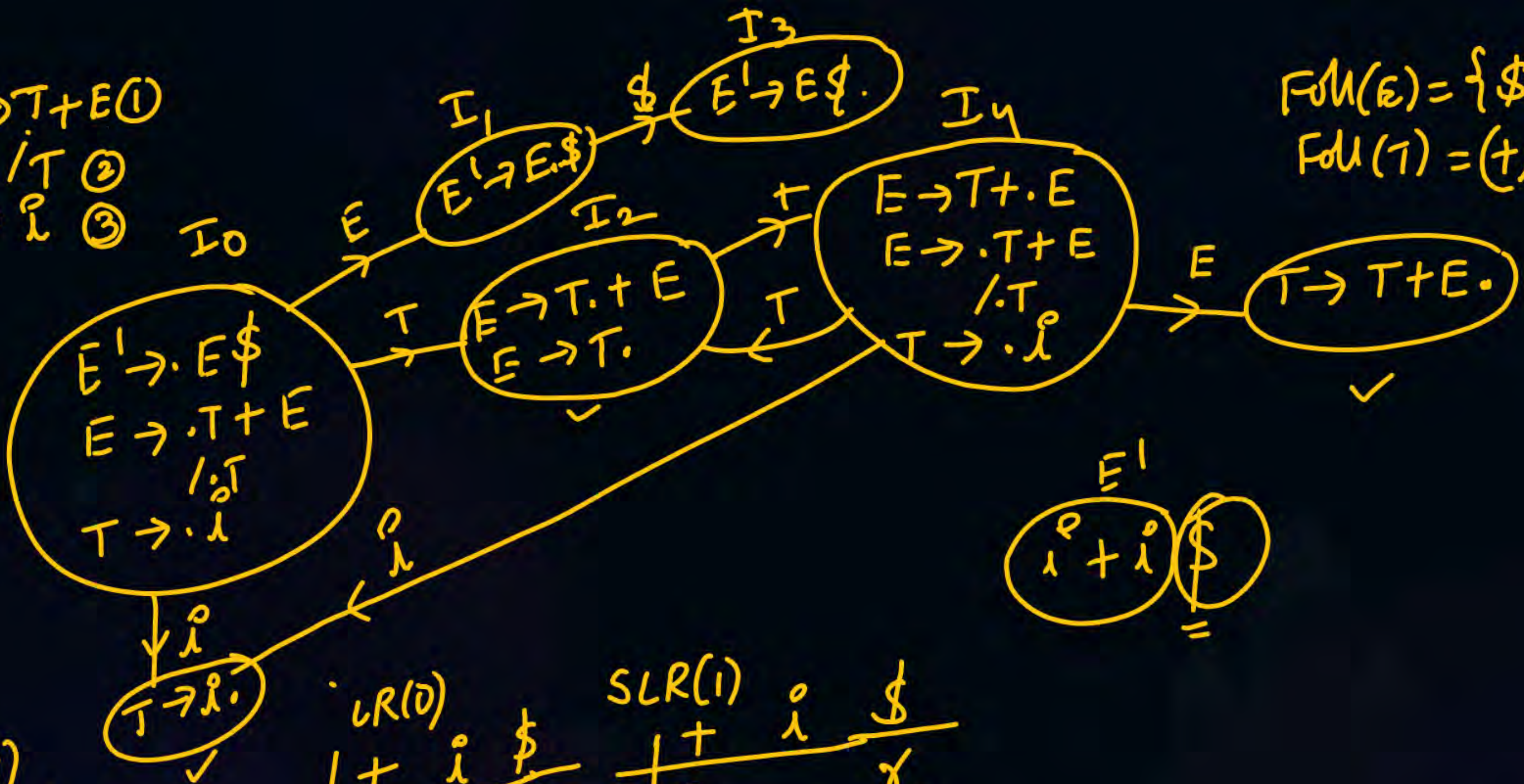
	+	i	\$
E		$E \rightarrow T + E /$ $E \rightarrow T$	

Not LL(1)

LL(1)
 LR(0)
 SLR(1)

$E \rightarrow T + E$ ①
 $\downarrow T$ ②
 $T \rightarrow \epsilon$ ③

$FOLLOW(E) = \{ \$ \}$
 $FOLLOW(T) = \{ +, \$ \}$



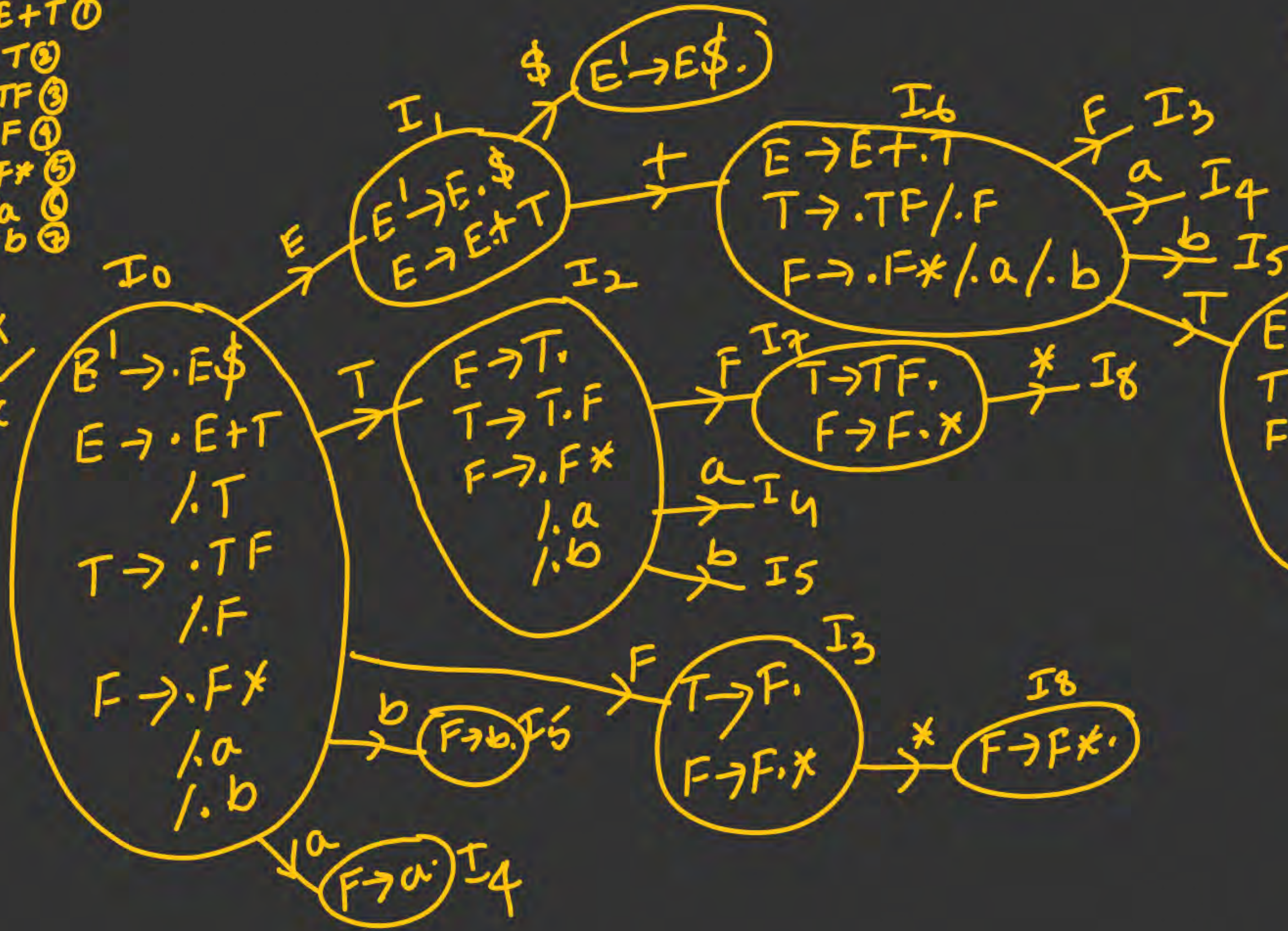
$LL(1)$
 $LR(0) \times$
 $SLR(1)$

LR(0)				SLR(1)			
	+	ϵ	\$		+	ϵ	\$
2	γ_2 / S	γ_2	γ_2	2	S		γ_2
				4			

$E \rightarrow E+T$ ①
 $\quad \quad \quad /T$ ②
 $T \rightarrow TF$ ③
 $\quad \quad \quad /F$ ④
 $F \rightarrow F*$ ⑤
 $\quad \quad \quad /a$ ⑥
 $\quad \quad \quad /b$ ⑦

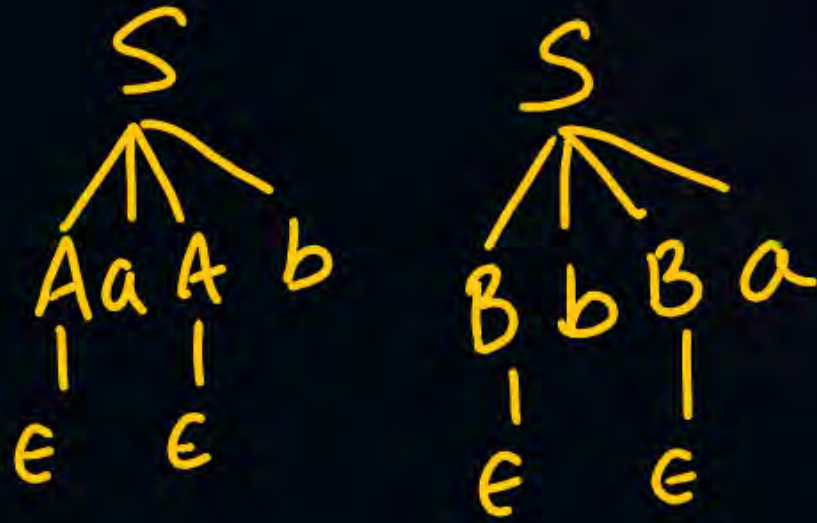
$Follow(E) = \{+, \$\}$ /
 $Follow(T) = \{+, \$, a, b\}$
 $Follow(F) = \{*, +, \$, a, b\}$

LL(1) X
 SLR(1) ✓
 LR(0) X



	a	b	+	\$	*
2	S4	S5	r2	r2	
3	r4	r4	r4	r4	S8
7	r3	r3	r3	r3	S6
9	S4	S5	r1	r1	

$S \rightarrow AaAb$ ①
 $S \rightarrow BbBa$ ②
 $A \rightarrow \epsilon$ ③
 $B \rightarrow \epsilon$ ④



$I = \{ab, ba\}$

$LL(1)$
 $LR(0)$
 $SLR(1)$

$LL(1)$

	a	b	\$
S	$S \rightarrow AaAb$	$S \rightarrow BbBa$	
A	$A \rightarrow \epsilon$	$A \rightarrow \epsilon$	
B	$B \rightarrow \epsilon$	$B \rightarrow \epsilon$	

$LL(1)$

SLR(1)

$S \rightarrow AaAb$ ①
 $\quad \quad \quad BbBa$ ②

$A \rightarrow \epsilon$ ③

$B \rightarrow \epsilon$ ④

I_0

$S' \rightarrow \cdot S$

$S \rightarrow \cdot AaAb$
 $\quad \quad \quad BbBa$

$A \rightarrow \cdot \cdot$

$B \rightarrow \cdot \cdot$

	a	b	\$
0	δ_3/δ_4	δ_3/δ_4	

$\cdot \epsilon$

$\epsilon \cdot$

$\cdot$

LL(1) ✓
 LR(0) ✗
 SLR(1) ✗

$S \rightarrow \underline{A}S / \underline{b}$
 $A \rightarrow SA / \underline{a}$

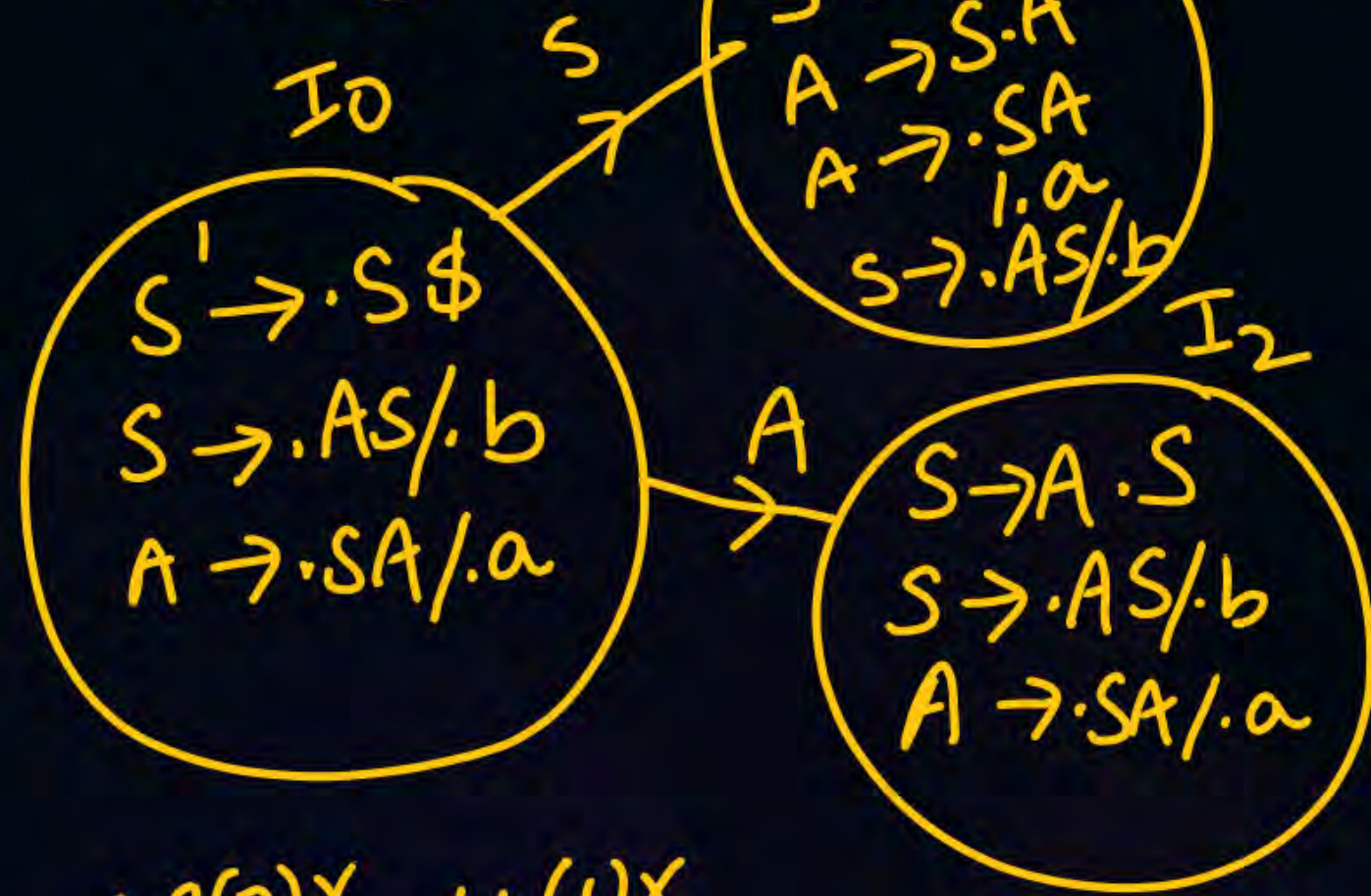
	LL(1)		
	a	b	\$
S	$S \rightarrow AS$	$S \rightarrow AS / S \rightarrow b$	
A			

$LL(1) \times$
 $LR(0)$
 $SLR(1)$

$\text{First}(A) = \{a, b\}$

5 min

①
 $S \rightarrow AS/b$ ②
 $A \rightarrow SA/a$
 ③ ④

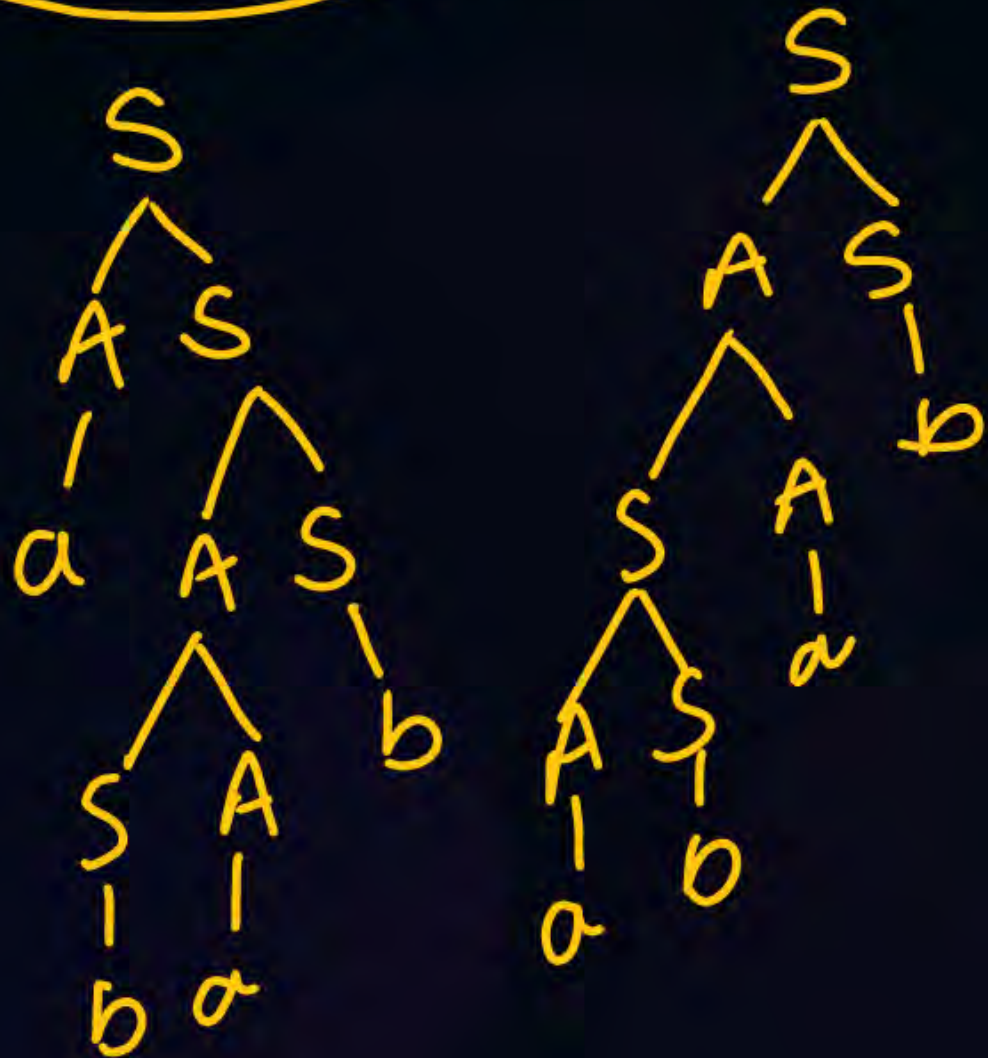
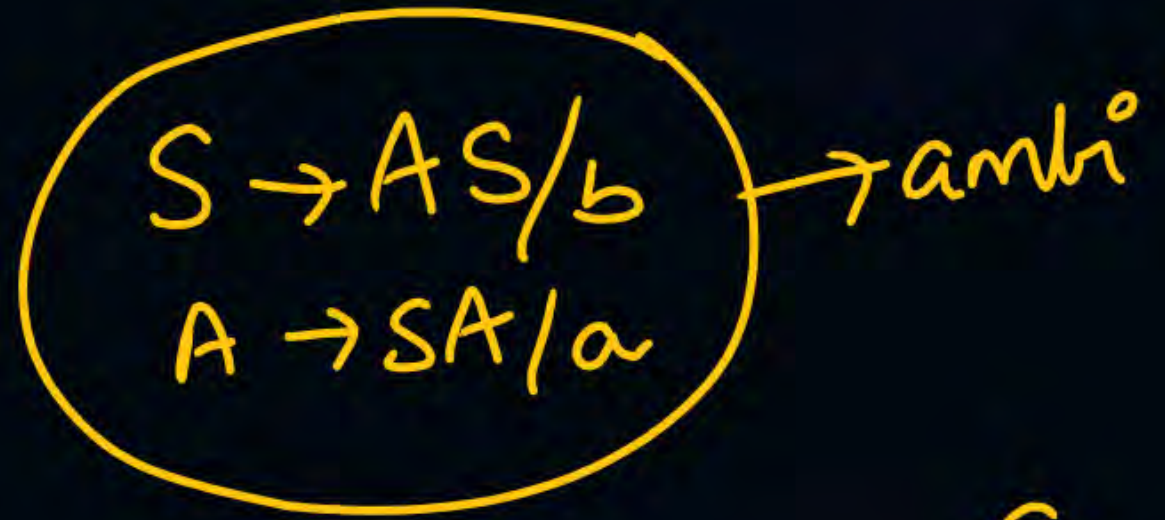


LR(0)X LL(1)X
 SLR1X

✓ SLR(1)

	a	b	\$
3		3/4	

$FOLLOW(S) = \{\$, a, b\}$
 $FOLLOW(A) = \{b, \}$



$S \rightarrow Aa \{d\}$
 $\quad \quad \quad /bAC \{b\}$
 $\quad \quad \quad /dC \{d\}$
 $\quad \quad \quad /bda$

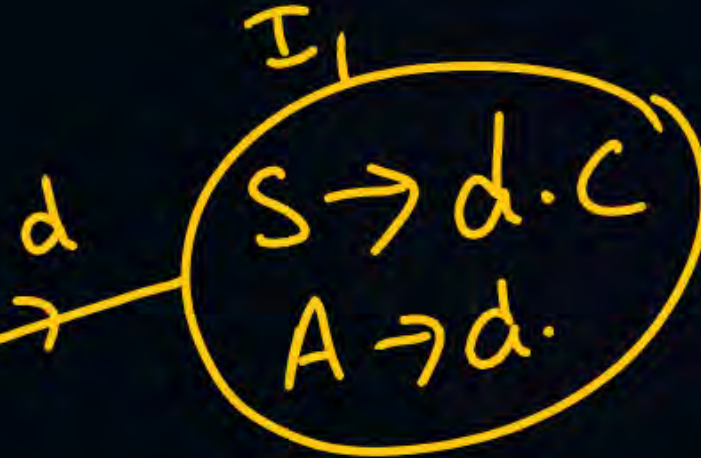
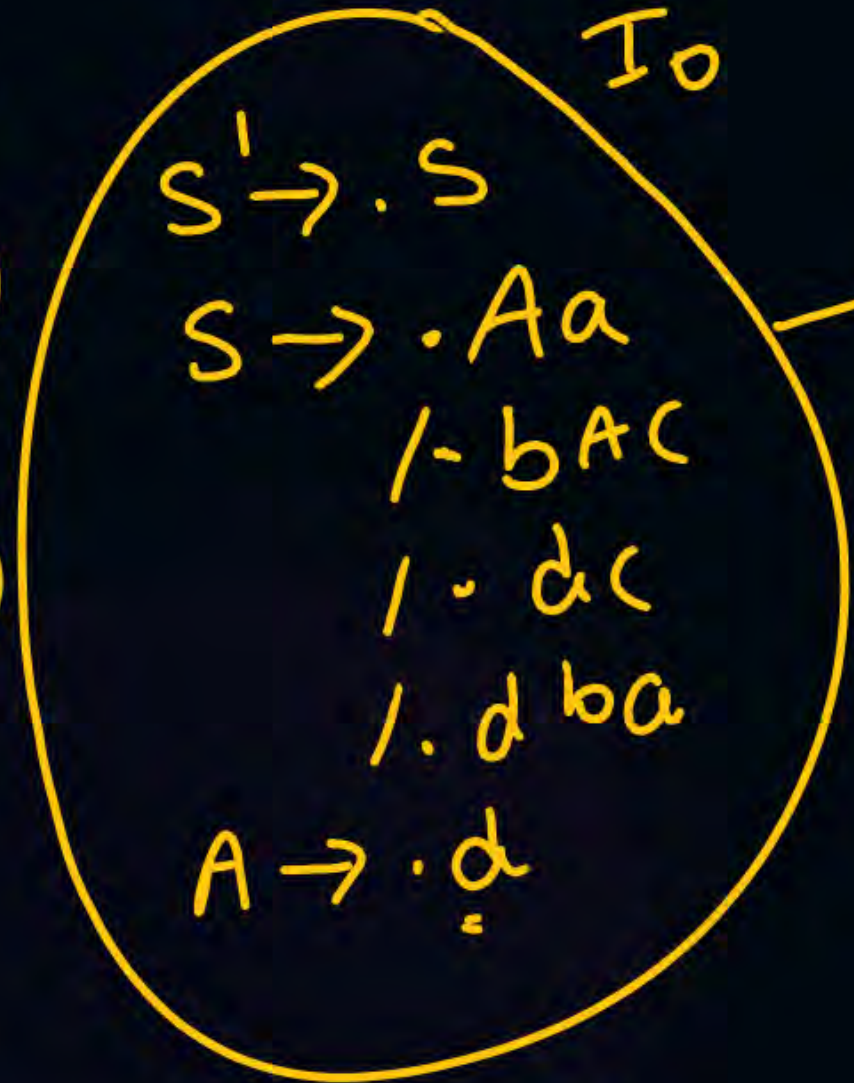
$A \rightarrow d$

$SLR(1)$
 $LL(1) \checkmark$
 $LR(0)$

$LL(1)$

	a	b	c	d	\$
S	:			$S \rightarrow Aa$ $\quad \quad \quad /S \rightarrow dC$	

$S \rightarrow A \underline{a}$ ①
 $/bA\underline{c}$ ②
 $/d\underline{c}$ ③
 $/bda$ ④
 $A \rightarrow d$ ⑤



	a	b	c	d	\$
1	δ_5		δ_5 / δ_2		

LR(0) X

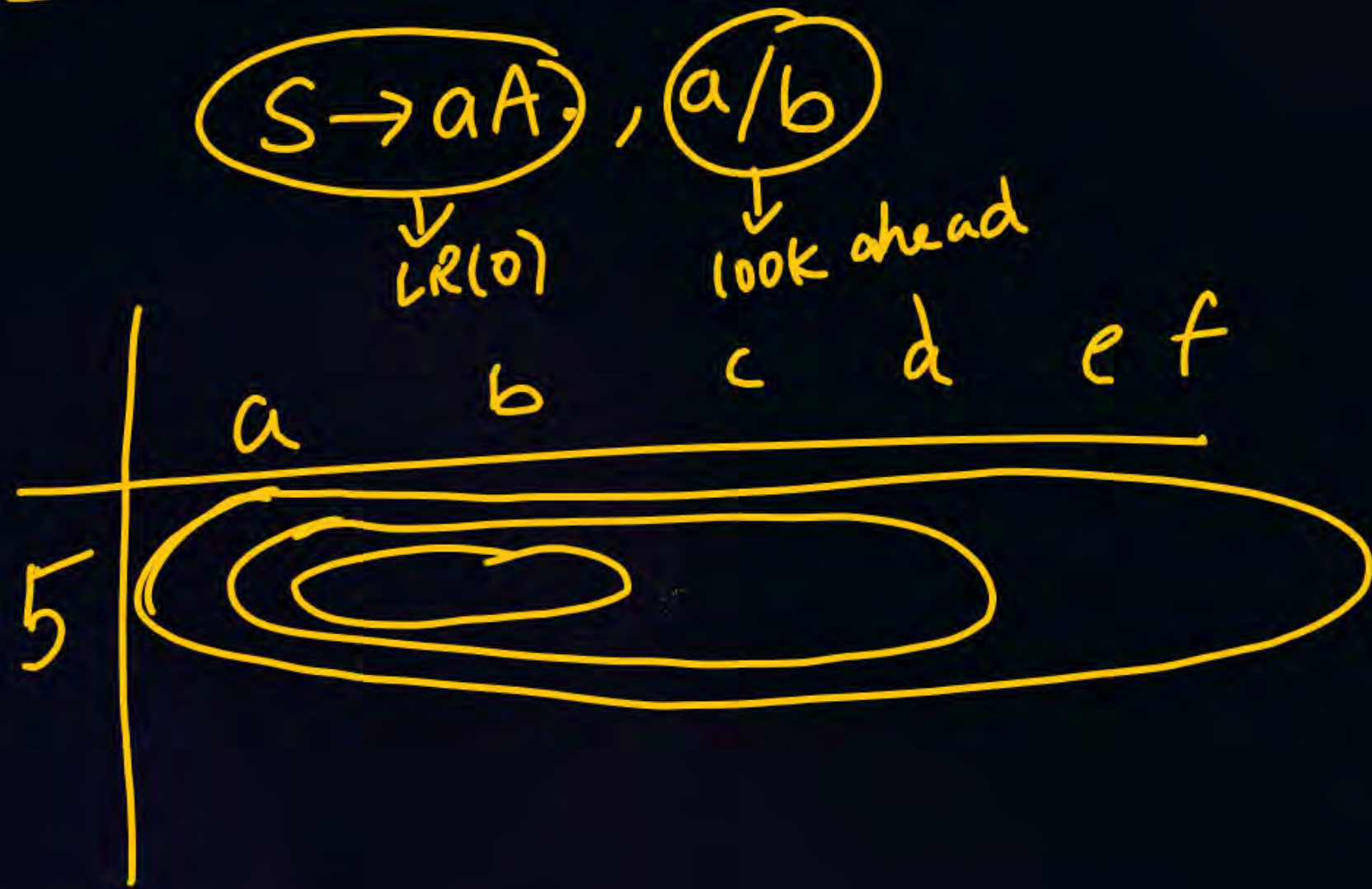
SLR(1) X

$F_{\text{FOLLOW}}(A) = \{c, a\}$

CLR(1) Δ LR(1) and LALR(1)

Among all the parsers CLR(1) is the most powerful parser
Parsing algorithm is same for all LR parsers
 only parsing tables change. In LR(0) reduce moves
 are placed in the entire row. In SLR(1) reduce moves are placed
 in follow (LHS). In CLR(1) and LALR(1) reduce moves are placed
 in the lookaheads. To construct CLR(1) and LALR(1),
 we use LR(1) items ✓

LR(1) item = LR(0) item + lookahead



Closure(I) where I is a set of LR(1) items :-

I is a Set of



THANK - YOU